

- A. A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight bit words.
1. How many bits are required for addressing the main memory? 22
 2. How many bits are needed to represent the TAG, SET and WORD fields? 9-5-8

Solution:

Given,

For main memory,

Number of blocks = 16,384

Number of 8-bit words per block = 256

For cache memory,

4-block set associative mapping is used

Number of 8-bit words per block = 256

Number of blocks = 128

Total addressing bits required = $\log_2 16384 + \log_2 256 = 14 + 8 = 22$

Total number of sets = $\frac{128}{4} = 32$

Bits required to represent WORD field = $\log_2 256 = 8$

Bits required to represent SET field = $\log_2 32 = 5$

Bits required to represent TAG field = $22 - 5 - 8 = 9$

- B. A computer has a 256 KB, 4-way set associative, write back data cache with block size of 32 bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.
1. The number of bits in the tag field of an address is? 16
 2. The size of the cache tag directory is? 20kB

Solution:

Given,

For main memory,

32 bit address

Size of each block = 32 bytes

For cache memory,

4-way set associative cache mapping is used

Total size of cache memory = 256KB

Size of each block = 32 bytes

Cache tag directory contains address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

$$\text{Total numbers of blocks} = \frac{256\text{KB}}{32\text{B}} = 8\text{K}$$

$$\text{Total numbers of sets} = \frac{8\text{K}}{4} = 2\text{K}$$

$$\text{Bits required to represent WORD field} = \log_2 32 = 5$$

$$\text{Bits required to represent SET field} = \log_2 2\text{K} = 11$$

$$\text{Bits required to represent TAG field} = 32 - 5 - 11 = 16$$

$$\text{Total numbers of bits per cache tag directory} = 16 + 2 + 1 + 1 = 20$$

$$\text{Total size of cache directory} = 8\text{K} \times 20 \text{ bits} = 20\text{KB}$$

- C. A 4-way set associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____. 20

Solution:

Given,

For main memory,

Total size of main memory = 4 GB

Size of each word = 32 bits

Total number of words in each blocks = 8

For cache memory,

4-way set associative cache memory is used

Total size cache memory = 16KB

Size of each word = 32 bits

Total number of words in each blocks = 8

$$\text{Total number of blocks} = \frac{16 \times 1024 \times 8}{8 \times 32} = 512$$

$$\text{Total number of sets} = \frac{512}{4} = 128$$

$$\text{Total addressing bits required} = \log_2 \frac{4 \times 1024 \times 1024 \times 1024 \times 8}{32} = 30$$

$$\text{Bits required to represent WORD field} = \log_2 8 = 3$$

$$\text{Bits required to represent SET field} = \log_2 128 = 7$$

$$\text{Bits required to represent TAG field} = 30 - 3 - 7 = 20$$

- D. If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?
1. Width of tag comparator
 2. Width of set index decoder
 3. Width of way selection multiplexer
 4. Width of processor to main memory data bus

4th option is right answer. Find yourself why it is.

- E. Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____. 18

Solution:

Given,

For main memory,

Total number of addressing bits = 32

Total size of main memory = 2^{32} bytes

Size of each block = 32 bytes

For cache memory,

Direct cache mapping is used

Total numbers of cache lines = 512

Size of each cache line = 32 bytes

$$\text{Bits required to represent WORD field} = \log_2 32 = 5$$

$$\text{Bits required to represent LINE field} = \log_2 512 = 9$$

$$\text{Bits required to represent TAG field} = 32 - 5 - 9 = 18$$

- F. Consider a system with main memory consisting of 8K blocks, a cache memory consisting of 256 blocks and a block size of 16 words. What will be the word field, block field and tag field length? How many bits are there in main memory address? If the system uses direct mapping and associative mapping techniques.

Solution,
Given,

For main memory,
Total number of blocks = 8K
Number of words in each block = 16

For cache memory,
Total number of blocks = 256
Number of words in each block = 16

$$\text{Total addressing bits required} = \log_2 8 \times 1024 \times 16 = 17$$

$$\text{Bits required to represent WORD field} = \log_2 16 = 4$$

For direct mapping,

$$\text{Bits required to represent LINE field} = \log_2 256 = 8$$

$$\text{Bits required to represent TAG field} = 17 - 8 - 4 = 5$$

For full associative mapping,

$$\text{Bits required to represent TAG field} = 17 - 4 = 13$$

For 4-way set associative mapping,

$$\text{Total number of sets} = \frac{256}{4} = 64$$

$$\text{Bits required to represent SET field} = \log_2 64 = 6$$

$$\text{Bits required to represent TAG field} = 17 - 6 - 4 = 7$$